13.9. NATURAL RADIOACTIVITY OF THE OCEANS

The total amounts of ^{238}U and ^{232}Th in the ocean are 4.3×10^{12} kg (53 EBq) and 6.9×10^{10} kg (0.3 EBq), respectively. The oceans contain a much higher concentration of uranium than would be expected from its abundance in rocks. It also occurs at much larger concentration than thorium, even though the latter element is more abundant in rocks. This is explained by the absence of easily soluble thorium complexes, in contrast to the strong $UO_2(CO_3)_3^{4-}$ complex. ^{238}U decays to ^{234}Th ($t_{½}$ 24.1 d), which precipitates out or absorbs to colloids before it decays to ^{234}U, causing a slight disequilibrium between the two U-isotopes. This is used to determine the mean residence time of Th in the oceans: ~200 y; by contrast, the residence time of U in the ocean is 5×10^5 y. The decay passes the long-lived ^{230}Th ($t_{½}$ 75 400 y), which deposits in sediments before radioactive equilibrium is reached with ^{226}Ra (t 1600 y); consequently the specific activity of ^{226}Ra is less than expected from the U-content (39 Bq/m^3), only 1–10 Bq/m^3. It should be noted that the ^{222}Rn concentration in surface waters is in disequilibrium with ^{226}Ra because Rn is vented by heat, turbulence, etc.

As the total content of ^{40}K in the oceans is 7.4×10^{13} kg, corresponding to 1.94×10^4 EBq, ^{40}K is the largest source of radioactivity in the oceans. Minor activities come from ^3H as HTO and from ^{14}C as dissolved CO_2 or HCO_3^-.

13.10. ANTHROPOGENIC RADIOACTIVITY IN NATURE

In the analysis of a sample for its content of natural radioactivity it is necessary today to consider the possibility that the sample has become contaminated by "non-natural" radioactivities, i.e. radionuclides added to by human activities (so-called *anthropogenic sources*). Nuclear weapons tests, nuclear satellites burnt-up in the atmosphere, and nuclear power accidents may release large amounts of activities, see Table 13.4. The nuclear power industry is permitted by health authorities to continually release small, controlled amounts of specified radionuclides into the atmosphere and into open waters, Tables 19.7 and 21.10. Effluents may also come from large radiotracer users (especially

Table 13.4 Events leading to large injections of radionuclides into the atmosphere

Source	Country	Time	Radioactivity (Bq)	Important nuclides
Hiroshima & Nagasaki	Japan	1945	4×10^{16}	Fiss.prod., Actinides
Atmospheric weapons tests	USA USSR	-1963	2×10^{20}	Fiss. prod., Actinides
Windscale	UK	1957	1×10^{15}	^{131}I
Chelyabinsk (Kysthym)	USSR	1957	8×10^{16}	Fiss. prod. ^{90}Sr, ^{137}Cs
Harrisburg	USA	1979	1×10^{12}	Noble gases, ^{131}I
Chernobyl	USSR	1986	2×10^{18}	^{137}Cs
Fukushima	Japan	2010	2×10^{18}	^{131}I, ^{137}Cs

hospitals). In most cases these releases are well known and the area concerned ("contaminated") identified. From a global standpoint, compared to natural radioactivity, these releases are minor, but they do add to the our "natural" radiation background (also called eco radiation), as discussed in Chapters 15 and 22.

Usually one distinguishes between "near field" and "far field" effects of radioactivity releases. Near field effects are observed close to the release source, as for example the nuclear power plant or nuclear waste storage facility. The dissolution of nuclear waste by rain or ground water is a typical near field problem. As the source is known, it can be controlled and its environment monitored. If the radioactivity exceeds permitted levels, access to the contaminated area can be restricted. Far field effects involve the behavior of radionuclides which have spread out of such a restricted area, caused either by nuclear power accidents and weapons tests or by leakage from nuclear power plants.

In this paragraph we briefly describe some of the largest anthropogenic sources causing far field effects, i.e. nuclear weapons tests and nuclear power plant accidents. The cause of the releases is discussed in Chapter 19. Chapter 22 discusses both near and far field effects in further detail, particularly with regard to chemical properties: liquid releases from nuclear power plants, dissolution of solidified nuclear waste and of fall-out particles, migration in the environment, and possible consequences.

13.10.1. Nuclear weapons

Nuclear weapons were tested up to 1963, the year of the test ban, with total releases up to 2×10^{20} Bq fission products, as well as some lesser amounts of Pu isotopes; cf. Table 22.2. Most of the debris injected into the troposphere had a mean residence time of $\sim$30 d, causing fall-out mostly in the neighborhood of the test area. Some of the debris passed through the tropopause and entered the stratosphere where it was carried by winds around the globe at approximately the latitude of the release. The residence time in the stratosphere is 3–24 months depending on latitude (greatest at the equator). The average surface deposition is 2500 Bq/m^2 at 40–50°N, decreasing towards north an south ($<$800 at $>$70°N and at $<$20°N). The global integrated deposition density for ^{241}Pu is 440 Bq/m^2 and the air concentration is 0.8 mBq/m^3.

Because the atmospheric tests stopped in 1963, the fission products have now decayed for $\sim$50 y, leaving only ^{90}Sr, ^{137}Cs and Pu "active". Comparing with global natural radioactivity the contributions of these isotopes to our environment is negligible today, except for T and ^{14}C.

13.10.2. Nuclear power plant accidents

In 1957 a fire developed in one of the gas cooled graphite moderated rectors at *Windscale*, UK. The main radionuclides released were (in TBq) ^{131}I 700, ^{137}Cs 20, ^{89}Sr 3 and ^{90}Sr

0.3. The largest air concentration close to the plant was 20 kBq/m^3. The Cs activity deposited was $\leq$4 kBq/m^2. The ground activity is now back to normal.

Also in 1957 an explosion occurred in a nuclear waste storage facility (probably due to reactions between organic matter and nitrate) at Kyshtym south of Sverdlovsk, former USSR, leading to the contamination of approximately 1600 km^2 of land by 8×10^{16} Bq fission products, causing local contaminations exceeding 10^{10} Bq/m^2 (2×10^8 Bq/m^2 ^{90}Sr and ^{137}Cs). The area is still uninhabitable, see also §21.10.

In 1979 a partial core melt-down occurred in a reactor at Three Mile Island power station (TMI), *Harrisburg*, Pennsylvania. Although the fission products released from the core were almost completely contained within the building, the Xe and Kr and some iodine ($\sim$1 TBq ^{131}I) were released. No deposition occurred on ground outside the building.

The explosion and fire in one of the power reactors at *Chernobyl*, then Soviet Union — now Ukraine, in 1986, was a more severe accident (§§20.1.2 and 22.3). For several days large amounts of fission products and actinides were ejected and spread over large areas of the former USSR and Europe, see Figure 13.10. Almost 20% of the

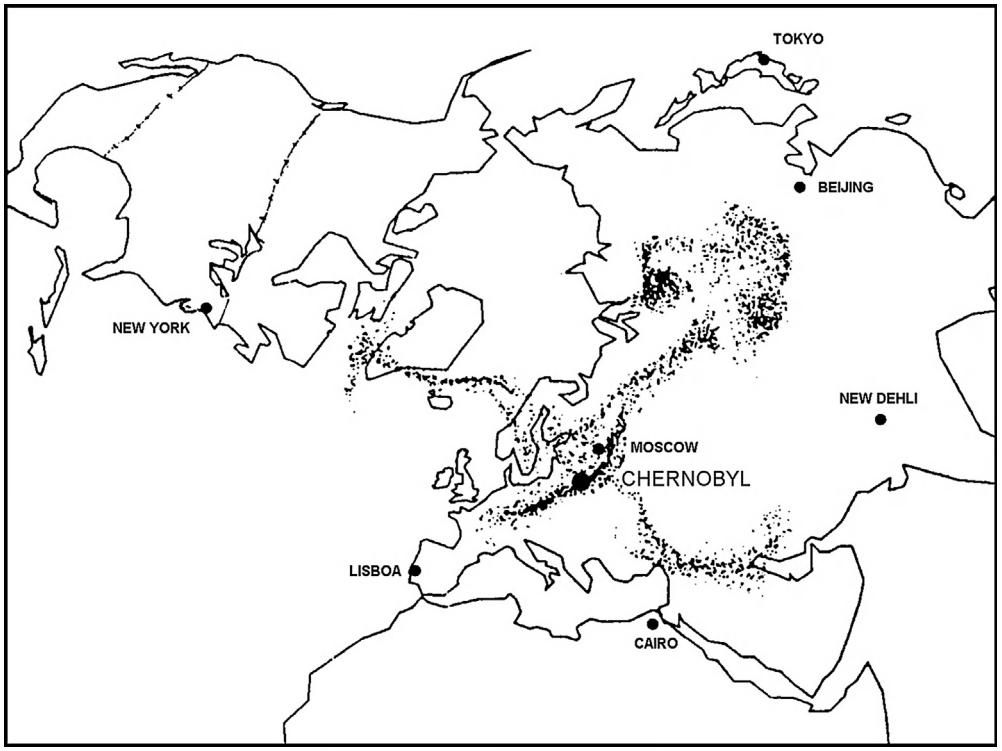


Figure 13.10 The distribution of the airborne radioactivity from Chernobyl.

fission products came down over Scandinavia, causing a deposition >120 kBq ^{137}Cs/m^2 over the city Gävle in Sweden (170 km north of Stockholm). The plume spread down over Central Europe, causing very uneven deposition due to wind pattern and local rains: e.g., Munich, Germany, received ~25 kBq ^{137}Cs and 0.2 kBq ^{90}Sr per m^2, while Mainz (400 km away) received 180 Bq ^{137}Cs and ~0.001 Bq ^{90}Sr per m^2, see also §22.3.

On the 11th of March 2011 an earthquake of the magnitude 9.0 on the Richter scale occurred outside the eastern coast of Japan. All affected nuclear reactors shut down as planned as a response to the earthquake. It should be noted that some of the reactors only had a design basis of 8.2 but handled the larger magnitude anyway. However, as a result of the earthquake a tsunami was also formed. As it reached the Japanese coast the devastation was huge resulting in more than 20 000 casualties. Some of the nuclear reactors were also affected and most severely where the ones at the Fukushima Daiichi plant. Even if the reactors handled the unusually strong earthquake the protection against a tsunami of this size was not good enough. The wall designed to protect the reactor from tsunamis was about 7 meters tall but the wave was more than 10 meters high as it hit the coast. The water masses affected the power supply and also the possibility to run the on site emergency diesel generators. Thus all six reactors suffered a total station blackout when the battery power was lost after a few hours. With a total loss of power the auxiliary cooling systems stopped and cooling water was lost to the reactor cores which were heated up due to the remaining decay heat in the fuel. However, the chain reactions had stopped before this point. As the fuel heated up the zircaloy cladding underwent a highly exothermic oxidation reaction according to (19.44), where T is in °C.

$$\mathrm{Zr} + 2\mathrm{H_2O} \rightarrow \mathrm{ZrO_2} + 2\mathrm{H_2} \quad \Delta H = -6.67 + 2.57 \times 10^{-4}\,\mathrm{T}\ (\mathrm{MJ/kg\ Zr}) \quad (19.44)$$

The hydrogen produced would normally have been converted to water by the installed electrically heated recombiners or vented by the fans to the stack. However, nothing of this worked after the total power loss. Thus hydrogen escaped up into the secondary reactor building where it accumulated and later exploded. The explosion did not affect the primary containment. However, the spent fuel storage on unit 4 was damaged and the stored fuel could no longer be cooled. That increased the risk for damage not only to the fuel inside the reactor cores but also to the spent fuel in the storage pond, which was not inside an effective containment. This increased the risk for another large release of radioactive material.

The release of radioactive material from the Fukushima accident was modest when compared to Chernobyl, ~5200 PBq total from Chernobyl and ~900 PBq total into the air from Fukushima. The release at Fukushima contained little, or no, corium and consisted mainly of the volatile fission products iodine and cesium (probably in the form

of CsOH, CsI, and I_2). However, significant land masses were contaminated. In most areas the contamination levels were comparable to those seen in many areas (e.g close to Gävle) in Sweden after the Chernobyl accident (about 120 kBq/m^2) where no evacuation or cleaning up measures were deemed necessary. However in the plume north of the plant in Fukushima there were areas with about 3 MBq/m^2.

An interesting observation from the Fukushima accident is the release pattern and the effect of weather at the time of the release. Most of the released activity ended up in the ocean due to the prevailing wind direction. However, over time the wind changed direction and the release was then spred over land. In retrospect it is now possible to see where it was raining at the time since CsOH and CsI are both water soluble and were washed out by the rain while some the elemental iodine (or iodine oxides) was transported through the precipitation zones.

13.10.3. Releases from nuclear power plants

All nuclear power plants are carefully controlled by the national radiation protection boards, and strict limits for releases are set. Usually it is quite easy for the power plants to meet these requirements. These releases are small compared to the natural radioactivities and mostly negligible in the far field, see §§20.4, 21.8, and 21.11.

13.10.4. Other anthropogenic additions

In 1975 the *London Dumping Convention* set limits to dumping of nuclear waste in the Oceans to places far of shipping trades and fishing areas and at depths >4000 m. Earlier dumping had been common practice by the nuclear powers even in narrow (the British Channel) and shallow (up to 50 m) waters to get rid of large volumes of low level long-lived waste. Officially reported dumpings were made as early as in 1946 and continued in the Pacific and the North Atlantic until 1982 and in the Barents and Kara seas even later. East of the British Islands the U.K. has dumped 665 TBq, France 134 TBq and other countries together some 35 GBq. In total some 45 PBq have been dumped at 46 different sites, mostly at depths of 1400–6500 m. Measurements reveal that in some cases radioactive nuclides are leaking out from the containers, causing bottom contamination at the site; however, the radionuclides are rapidly diluted as they are carried away by bottom currents.

In the 1990ies it was disclosed that the former USSR made huge dumps of nuclear waste in the waters east of Novaya Zemlya: some 10 000 containers and 13 nuclear reactors (8 still containing fuel), mainly from nuclear submarines. In some places the waste is at very shallow waters (some 30 m). The total activity is estimated to >60 PBq. The waste will ultimately leak out and spread with westerly currents towards the Barents Sea and the North Atlantic. However, the releases are expected to be rapidly diluted to harmless concentrations.

13.11. EXERCISES

13.1. What proof exists that some cosmic rays do not come from the sun?

13.2. (a) What is the primary cosmic radiation hitting earth's atmosphere? (b) Does it penetrate to the earth's surface?

13.3. What background from cosmic radiation is expected for an unshielded 100 ml ion chamber which exhibits an area of 100 cm^2 perpendicular to the direction of the cosmic radiation?

13.4. Which type of mesons are released in high energy particle interactions, and why?

13.5. Cosmic-ray irradiation of the atmosphere yields 0.036 ^{10}Be atoms cm^{-2} s^{-1}. If this ^{10}Be is rapidly carried down into seawater, which is assumed to have a volume of 1.4×10^{18} m^3, what will the equilibrium radioactivity of ^{10}Be in 1 m^3 seawater be? The earth's total surface area is 510×10^6 km^2.

13.6. In Greenland ice the ^{10}Be radioactivity has been measured to be 0.0184 dpm m^{-3}. How old is this ice if it was formed out of water in equilibrium with cosmic-ray ^{10}Be (see previous question)?

13.7. The CO_2 in the atmosphere is in exchange with carbon in living organisms, humus, dissolved organic compounds, and carbonate in the sea water, the latter being the main reservoir (88%) of all exchangeable carbon. The amount of such exchangeable carbon is estimated to be 7.9 g cm^{-2} of the earth. When cosmic-ray produced ^{14}C is mixed into this exchangeable carbon, what will the specific activity become?

13.8. 0.11 cm^3 helium gas at NTP was isolated from 100 g of uranium mineral containing 5 ppm uranium. How old is the mineral?

13.9. A mineral was found to contain 39.1 g K and 87.2×10^{-6} liter Ar at NTP. How old is the mineral?

13.10. A uranium mineral was found to contain the lead isotopes ^{204}Pb, ^{206}Pb, and ^{207}Pb in the ratio 1:1087:388, as determined with a mass spectrometer. Estimate the age of the mineral.

13.11. The heat flow from the earth's crust is 0.060 W m^{-2}. The mean thickness of the crust is 17 km and the earth's radius is 6371 km. The average concentration of uranium, thorium, and potassium in granite is estimated to be 4 ppm (by weight) 18 ppm, and 3.6%, respectively. Assuming that 7% by volume of the crust is made up of granite (feldspar + quartz, density 2.6 g cm^{-3}), what will the heat flow at the earth's surface be from each of these elements? Assume β-heat as 1/3 E_{max}; for α-decay assume $E_\alpha = Q_\alpha$. Each ^{40}K decay by the EC-branch emits one 1.46 MeV γ. Discuss the results.

13.12. A 1 GW_e nuclear power station uses annually about 30 t uranium enriched to 3% ^{235}U. (a) How much natural uranium consumed each year to keep it running? Assume waste stream from isotope separation plant to contain 0.3% ^{235}U.